

Question:

Carla is downloading a 200 GB file. Normally she can download 2 GB/minute, but 40% of the way through the download, Windows forces a restart to install updates, which takes 20 minutes. Then Carla has to restart the download from the beginning. How long does it take to download the file?

Reasoning + Answer:

Let's go step by step .

Step 1 : Understand the original scenario .
Carla normally downloads at 2 GB per minute .
The file is 200 GB .

If there were no interruptions , the time would be :
Total time = File size / Speed = 200 GB / 2 GB/min = 100 minutes .

But there is an interruption .

Step 2 : When does the interruption happen ?
" 40 % of the way through the download " â that means after downloading 40 % of 200 GB .

40 % of 200 GB = 0.4 Ã 200 = 80 GB .

So , she downloads 80 GB at 2 GB / minute .

Time to download 80 GB = 80 / 2 = 40 minutes .

Step 3 : Windows forces a restart to install updates â takes 20 minutes .
So , during that 20 minutes , nothing is downloading .

Then she restarts the download from the beginning â so she has to go back to 0 GB and start over .

Now , it's important to realize :
- From 0 to 80 GB again (the same 80 GB is downloaded again),
at 2 GB/min â takes 80 / 2 = 40 minutes .

But after the 20 minutes of update install , the download continues from scratch .

So total time :
= time for first 80 GB (40 min)
+ update time (20 min)
+ time to download the full file again (from 0 to 200 GB)
But wait â she has to re-download the entire file from 0 again ?

But no . The system restarts the download from the beginning (0), so she needs to get the whole 200 GB again â from the beginning .

She downloads the first 80 GB (40 min), then the 20-min update , then she restarts and has to download the entire file (200 GB) again , from scratch .

Therefore :
- After the restart , the download must begin again from 0 .
- So the total time includes the first 80 GB (40 min), plus 20 minutes for the update , plus the full 200 GB from scratch .

But from scratch again , at 2 GB/min â 200 / 2 = 100 minutes .

So total time :
= 40 (first part) + 20 (update) + 100 (full restart from beginning) = 160 minutes .

Wait â but is that correct ?

Another interpretation : Is "restart from the beginning" meaning only the download from 0 to 80 again ?

But that does not make sense â if the download fails at 80 GB , and restarts from the beginning , then she must download 200 GB from 0 again .

But no , the system may not just restart at 80 GB . It restarts from the beginning (0).

Therefore , it has to be from 0 again .

So the entire file must be re-downloaded â 200 GB from 0 â at 2 GB/min .

So final steps :

- Time to download first 80 GB = 80 / 2 = 40 minutes
- Update install : 20 minutes
- Then restart â now download from 0 again â takes 200 / 2 = 100 minutes

Total time = 40 + 20 + 100 = 160 minutes .

Alternatively : It could be interpreted that the restarts go only from the start to 80 , meaning she only starts the download again from 0 to 80 , and then continues from there ?

But no â the problem says : "Windows forces a restart to install updates , which takes 20 minutes . Then Carla